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Optimality of central and projection algorithms for bounded uncertainty *

B.Z. KACEWICZ **

Dipartimento di Automatica e Informatica, Politecnico di Torino, Corsa Duca degli Abruzzi 24, 10129 Torino, Italy

M. MILANESE

Dipartimento di Automatica e Informatica, Politecnico di Torino, Torino, Italy

R. TEMPO

Centro Elaborazione Numerale dei Segnali (CNR), Torino, Italy

A. VICINO

Dipartimento di Automatica e Informatica, Politecnico di Torino, Torino, Italy

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Abstract: This paper investigates optimality properties of central and projection algorithms for linear problems in the field of system identification in a context in which uncertainty is described in a deterministic rather than statistical way. Particular attention is devoted to least-squares algorithms when the measurement noise is assumed to be unknown but bounded in a Hilbert norm.

A major contribution of this paper consists in proving that least-squares algorithms enjoy strong optimality properties. On the contrary, it is pointed out that these properties do not hold for other frequently used projection algorithms, such as least-absolute-values or minimax algorithms, corresponding to a description of the measurement error in l_1 or l_∞ norm, respectively.

Keywords: Linear problems, System identification, Deterministic uncertainty, Projection algorithms, Least squares, Local errors, Strongly optimal algorithms.

1. Introduction

In this paper we study some optimality properties of projection algorithms in a general setting which includes problems such as system approximation, parameter estimation, time-series prediction, etc. The most popular examples of such algorithms are least-squares algorithms whose importance is well recognized in many research areas. These algorithms are essentially based on the minimization of a Hilbert norm of the residual. In recent years several different projection algorithms assuming other norm characterizations of the residual have been proposed and studied. For example, the use of an l_1 norm leads to a least-absolute-values criterion, while the use of an l_∞ norm leads to a minimax criterion [1,2].

Most optimality properties of least squares have been derived under statistical assumptions on the data. In situations where no reliable statistical information about the data is available, e.g. in the presence of unknown error correlations or modelling errors, optimality properties of least squares are not known a priori. However, practical experience has shown that such algorithms behave satisfactorily even if some of the standard assumptions are not met.

The aim of this paper is to investigate optimality properties of least-squares algorithms when uncertainty is described in a deterministic instead of statistical way. In particular, we assume that the measurement noise is unknown but bounded in a certain norm.

Our approach is founded on information-based complexity theory (see [3,4]). Thus theory has been successfully used to study optimal algorithms for solving numerical problems such as function approximation and integration, nonlinear equations, etc. Recently, this approach has also been used to develop and investigate optimal algorithms for solving estimation and prediction problems (see [5,6,7]).

Let us now outline the general setting of the problem. We are interested in estimating the image Sx of an unknown element x belonging to a finite-dimensional linear space X. We assume that the linear operator S mapping X into a finite-di-

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** On leave from Institute of Informatics, University of Warsaw, Warsaw, Poland.

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mensional linear normed space Z is known and that only corrupted information y about x is available. More precisely, information y belonging to a finite-dimensional linear normed space Y is of the form $y = Ix + \rho$, where I is a given linear information operator and ρ is a unknown noise bounded in the norm of the space Y. Estimates of Sx in the space Z are obtained by a suitable transformation $A(y)$ of the available data y. Such a transformation (possibly nonlinear) is called an algorithm. In this framework, a least-squares estimator is an example of an algorithm and the main objective of this paper is to investigate some of its optimality properties. Three types of errors (and the corresponding optimality concepts) are defined (see also [7]). The X-local error is the maximum error for a fixed x with respect to all possible y consistent with x; the Y-local error is the maximum error for a given corrupted information y with respect to all x consistent with y; the global error is the maximum error with respect to both x and y. It is important to notice that local errors are defined before are of particular interest in problems of system parameter estimation or time-series prediction. For what concerns the Y-local error, in these problems typically a measurement y is available and one wants to determine optimal estimates of Sx for each possible y using an algorithm A. On the other hand, the X-local error is also particularly meaningful as it measures the maximum uncertainty of the estimates due to all possible corrupted measurements y which may come out when observing Ix.

The concepts of strong and global optimalities are defined according to definitions of the corresponding local and global errors. An algorithm is called X-strongly optimal, Y-strongly optimal or (globally) optimal if it minimizes the X-local, Y-local or global error, respectively.

The main contribution of the paper consists in showing that if the space Y is equipped with a Hilbert norm, least-squares algorithms enjoy the following properties:

(a) they are X-strongly optimal in the class of those algorithms, called correct algorithms, which recover the exact solution Sx if information is exact;

(b) they are Y-strongly optimal.

The above properties explain the good behaviour of least-squares algorithms also in some practical situations where no statistical information is available.

On the other hand, the above properties (a), (b) do not hold for projection algorithms for a general norm in the space Y. In particular, it is shown that algorithms based on least-absolute-values or minimax criteria (corresponding to the use of an l_1 or l_∞ norm in the space Y, respectively) are neither X- or Y-strongly optimal nor (globally) optimal.

For general norms it is not known if there exist algorithms combining X- and Y-strong optimality. However, an algorithm exists, called a central algorithm, which is Y-strongly optimal [3,4]. It has been shown in [7] that central algorithms are also X-strongly optimal in the class of correct algorithms for l_∞ norms in Y and Z. We generalize these results and show the relation between least-squares algorithms and central algorithms.

Table 1

IF			THEN		
norm in Y	norm in Z	type A	X-strongly optimal	Y-strongly optimal	globally optimal
general	general	central algorithm	?	yes (by definition)	yes
l_∞	l_∞	central algorithm	yes (among correct algorithms)	yes	yes
l_2	general	l_2 projection (LS)	yes (among correct algorithms)	yes	yes
l_1	whatsoever	l_1 projection	no [a]	no	no
l_∞	whatsoever	l_∞ projection	no	no	no
a strictly convex norm s	whatsoever	s projection	no	no	no

[a] 'no' means 'certainly not generally true' as is proved by counter examples.

In Section 2 basic definitions and notations are briefly introduced. In particular, local and global errors of an algorithm and related optimality concepts are defined. Section 3 recalls the definition of a central algorithm and presents some of its properties. Section 4 contains the main results of the paper showing optimality properties of least-squares algorithms when a Hilbert norm is used in the space Y. Illustrative examples of the concepts introduced in previous sections are provided in Section 5. These examples also show that optimality properties do not hold when Y is equipped with l_1, l_∞ or certain strictly convex norms. The obtained results are summarized in Table 1.

2. Definitions and notations

This section provides formal definitions and notations used in te paper.

Let X be a finite-dimensional linear space over the real field. Let us consider a given linear operator S, called *solution operator*, mapping X into Z,

$$S: X \to Z, \tag{1}$$

where X is a finite-dimensional linear normed space over the real field. Our aim is to approximate an element Sx, knowing perturbed information on the problem element x.

Let a linear operator I, called *information operator*, transform X into a linear finite-dimensional normed space Y,

$$I: X \to Y. \tag{2}$$

We assume that $\dim Y \geqslant \dim X$ and that the information operator I is complete, i.e., it is a one-to-one mapping. The former assumption is satisfied in most practical problems that we are interested in, i.e., system-parameter estimation and time-series prediction. The latter assumption means that we can exactly recover x from the knowledge of exact information Ix. In general exact information Ix about x is not known but only approximate information y is available such that

$$\| y - Ix \| \leqslant \varepsilon \tag{3}$$

where ε is some fixed nonnegative number. This means that the information uncertainty $\rho = y - Ix$ is assumed unknown but bounded. From the previous assumptions it follows that the solution uncertainty is due only to the perturbation of the information.

An *algorithm* is a mapping (in general nonlinear) from Y into Z,

$$A: Y \to Z, \tag{4}$$

which provides an approximation $A(y)$ to Sx using approximate information y. Such an algorithm will often be called an *estimator* of the solution.

Let

$$V(x) = \{ y \in Y: \| y - Ix \| \leqslant \varepsilon \}. \tag{5}$$

We now define an *X-local error* $E_1(A, x)$ *of an algorithm* A for each element $x \in X$:

$$E_1(A, x) = \sup_{y \in V(x)} \| Sx - A(y) \|. \tag{6}$$

An algorithm A^* is called *X-strongly optimal* in a class $\mathscr{A}$ of algorithms if

$$\begin{aligned} &A^* \in \mathscr{A} \quad \text{and} \\ &E_1(A^*, x) \leqslant E_1(A, x) \quad \forall A \in \mathscr{A}, \forall x \in X. \end{aligned} \tag{7}$$

The X-strong optimality is a meaningful property in estimation problems as it ensures the minimum error of the estimates $A(y)$ for the worst measurement y belonging to the set $V(x)$, for any (unknown) element x.

We also define a *Y-local error* $E_2(A, y)$ *of an algorithm* A for each approximate information y. Let

$$T(y) = \{ x \in X: \| y - Ix \| \leqslant \varepsilon \} \tag{8}$$

and

$$Y_0 = \{ y \in Y: T(y) \neq \emptyset \}. \tag{9}$$

We define for $y \in Y_0$,

$$E_2(A, y) = \sup_{x \in T(y)} \| Sx - A(y) \|. \tag{10}$$

An algorithm $\hat{A}$ is called *Y-strongly optimal* in a class $\mathscr{A}$ of algorithms if

$$\begin{aligned} &\hat{A} \in \mathscr{A} \quad \text{and} \\ &E_2(\hat{A}, y) \leqslant E_2(A, y) \quad \forall A \in \mathscr{A}, \forall y \in Y_0. \end{aligned} \tag{11}$$

The Y-strong optimality is a particularly interesting property in estimation problems where we often wish to estimate Sx with the best achievable accuracy for the worst element x belonging to

$T(y)$, for any fixed (known) approximate information y.

An example of a class of algorithms $\mathscr{A}$, considered in estimation problems (see [8]), is given by the class $\mathscr{A}^c$ of correct algorithms. We say that an algorithm A is *correct* if

$$A(Ix) = Sx \quad \forall x \in X. \tag{12}$$

This means that a correct algorithm gives the exact solution if information is exact. It should be noticed that the concept of correct algorithm is meaningful only when the information operator is complete.

Let us now consider an example which explains the role that the various spaces and operators introduced above play in a simple parameter estimation problem.

Example 1. Suppose that the space X is the following one-dimensional space of real-time functions:

$$X = \{ f(\cdot) : f(t) = xt \ \forall x \in R \}. \tag{13}$$

This space can be also identified with the space of real numbers x ($X = \mathbb{R}$) because each value of the parameter x uniquely defines a time function and vice-versa. In the following we will refer to one of these equivalent interpretations according to the particular context.

We consider the problem of estimating the parameter x when two measurements of $f(t)$ are known not exactly, but with some norm-bounded uncertainty. In this context, the solution space Z is the space of the parameters x to be estimated and Y is the two-dimensional measurement space

$$Z = \mathbb{R}, \qquad Y = \mathbb{R}^2. \tag{14}$$

Thus, the solution operator $S : \mathbb{R} \to \mathbb{R}$ is the identity operator

$$Sx = x \tag{15}$$

and the information operator I is a sampling operator such that

$$Ix = [f(1), f(2)] = [x, 2x]. \tag{16}$$

The actual information belonging to the space Y and corrupted by noise is given by

$$y = Ix + \rho, \tag{17}$$

where the unknown two-dimensional error vector ρ is such that

$$\| \rho \| \leqslant \varepsilon, \tag{18}$$

a fixed nonnegative constant.

3. Optimality properties of central algorithms

We now recall the concept of a central algorithm A_c (see [3]). Let $y \in Y_0$ and $U(y) = S(T(y))$. By definition, a *central algorithm* A_c provides an approximation $A_c(y)$ which is the center of the minimal ball containing the set $U(y)$ (i.e., it is a center of the set $U(y)$ in the sense of Chebychev (see [9]). More precisely, $A_c(y)$ is an element of the space Z such that

$$\sup_{u \in U(y)} \| u - A_c(y) \| = \inf_{z \in Z} \sup_{u \in U(y)} \| u - z \|. \tag{19}$$

As the center of $U(y)$ need not be unique, we shall use some criterion for its choice to define $A_c(y)$ uniquely.

Example 2. We now show how to compute a central algorithm for the problem previously introduced in Example 1 of Section 2, under the assumption of l_∞ norm in the space Y.

In Figure 1, which gives a geometric representation of the problem, the line $y_2 = 2y_1$ is the range of the information operator. The interval AB represents the set $T(y)$ defined in (8). The set Y_0 defined in (9) is obtained by finding conditions under which the ball of fixed radius ε centered at $y = [y_1, y_2]$ intersects the range of I. Straightforward computations lead to the following result:

$$Y_0 = \{ y \in Y : 2y_1 - 3\varepsilon \leqslant y_2 \leqslant 2y_1 + 3\varepsilon \}. \tag{20}$$

Since S is identity, AB also represents the set $U(y)$. It follows from (19) that a central algorithm $A_c(y)$ is given by the center of the interval AB. In particular, if $2y_1 - 3\varepsilon \leqslant y_2 \leqslant 2y_1 - \varepsilon$, as in Figure 1, easy computations lead to

$$A_c(y) = \tfrac{1}{4}(2y_1 + y_2 - \varepsilon). \tag{21}$$

It is worthwhile to observe that in this case the center is unique.

Some optimality properties of central algorithms are now recalled. The following theorem

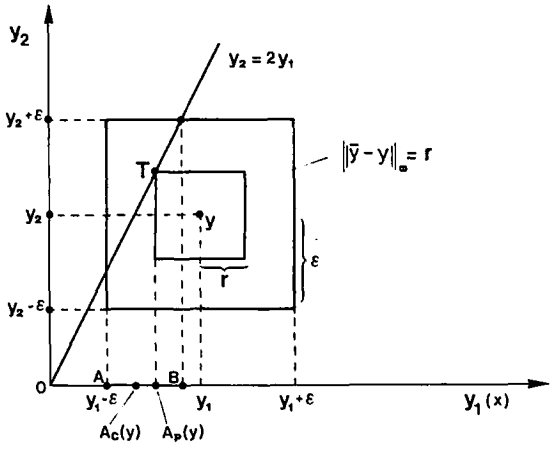

Fig. 1. A parameter estimation problem with l_∞ norm in the space Y.

states Y-strong optimality of central algorithms (see [4,9]).

Theorem 1. *For any S and I (not necessarily linear) any central algorithm A_c is Y-strongly optimal in the class of all algorithms* (4), *i.e.,*

$$E_2(A_c, y) \leqslant E_2(A, y) \quad \forall A, \forall y \in Y_0. \tag{22}$$

The proof of this theorem follows immediately from the definition of A_c.

In [7] the X-strong optimality properties of central algorithms have been investigated. It has been shown there that if the spaces Z and Y are equipped with l_∞ norms, then a correct central algorithm A_c is X-strongly optimal in the class of correct algorithms. It is not difficult to observe (see Theorem 2 below) that this property holds for any norm in Y. Furthermore, Theorem 2 shows that, for arbitrary norms in Y and Z, the X-local error of any central algorithm is at most twice larger than the error of any correct algorithm.

In order to prove Theorem 2 we recall some known results concerning global optimality of A_c. The *global error* $E(A)$ of an algorithm A is defined as (see [9])

$$E(A) = \sup_{x \in X} E_1(A, x) \left(= \sup_{y \in Y_0} E_2(A, y) \right). \tag{23}$$

Let us denote by E the *minimal global error* of an algorithm (also called *intrinsic error* of the problem)

$$E_\varepsilon = \inf_A E(A), \tag{24}$$

where the infimum is taken over all algorithms (4).

Then it follows from Theorem 1 that A_c is optimal in a global sense, i.e.,

$$E(A_c) = E_\varepsilon. \tag{25}$$

In [9] upper and lower bounds on E_ε are given by means of the following inequalities:

$$e(\varepsilon) \leqslant E_\varepsilon \leqslant 2e(\varepsilon), \tag{26}$$

where

$$e(\varepsilon) = \sup_{\|Ix\| \leqslant \varepsilon} \|Sx\|. \tag{27}$$

Moreover, if S is a linear functional (dim $Z = 1$) then (see [10])

$$E_\varepsilon = e(\varepsilon). \tag{28}$$

It can be easily observed that this equality also holds if dim $Z \geqslant 1$ and Z is equipped with l_∞ norm.

We are now ready to prove the theorem.

Theorem 2. (a) *If Z is equipped with l_∞ norm then for any norm in Y,*

$$E_1(A_c, x) \leqslant E_1(A, x) \quad \forall A \in \mathscr{A}^c, \forall x \in X. \tag{29}$$

(b) *For any norm in Y and Z, we have*

$$E_1(A_c, x) \leqslant 2E_1(A, x) \quad \forall A \in \mathscr{A}^c, \forall x \in X. \tag{30}$$

Proof. For any correct algorithm $A \in \mathscr{A}^c$ we have

$$E_1(A, x) = \sup_{y: \|y - Ix\| \leqslant \varepsilon} \|Sx - A(y)\| \geqslant \sup_{\hat{x}: \|I\hat{x} - Ix\| \leqslant \varepsilon} \|Sx - A(I\hat{x})\| = e(\varepsilon). \tag{31}$$

From the other side, using (23) and (25) we can see that

$$E_1(A_c, x) \leqslant E(A_c) = E_\varepsilon. \tag{32}$$

Thus, inequalities (26) and (31) yield

$$E_1(A_c, x) \leqslant E_\varepsilon \leqslant 2e(\varepsilon) \leqslant 2E_1(A, x) \quad \forall A \in \mathscr{A}^c, \tag{33}$$

which proves (30).

As we mentioned after relation (28), if the space Z is equipped with l_∞ norm then $E_\varepsilon = e(\varepsilon)$. Hence, from (31) and (32) we get for l_∞ norm in Z,

$$E_1(A_c, x) \leqslant E_1(A, x) \quad \forall x \in X, \forall A \in \mathscr{A}^c, \quad (34)$$

which proves relation (29). □

It is to remark that the statement (a) of Theorem 2 is still true even if Z is not equipped with l_∞ norm. In fact, the proof of the theorem remains true for all norms in Y and Z such that relation $E_\varepsilon = e(\varepsilon)$ is satisfied.

In the next section we shall investigate, in the light of the theoretical framework previously introduced, some optimality properties of the very well known and widely used class of least-squares estimators.

4. Properties of least-squares estimators

Optimality properties of least-squares estimators have been extensively studied in the literature in a statistical context. In this section we analyze their optimality properties in the context in which the uncertainty is supposed unknown but bounded (see Section 2). In particular we prove that least-squares estimators are X- and Y-strongly optimal.

In this section we assume that the space Y is equipped with a Hilbert norm $\|\cdot\|_H$ and we do not impose any restriction on the norm in the space Z.

Let us recall a definition of least-squares algorithm A_{LS}.

Definition 1. Let an element $x_y \in X$ be such that

$$\| Ix_y - y \|_H = \min_{x \in X} \| Ix - y \|_H. \quad (35)$$

Then the *least-squares algorithm* A_{LS} is defined as

$$A_{LS}(y) = Sx_y. \quad (36)$$

In other words an element Ix_y is a unique orthogonal projection of y on the range space of the information operator I. Since I is one-to-one, an element $x_y \in X$ can be uniquely recovered from information Ix_y, which means that the algorithm A_{LS} is well defined. Let P denote the projection operator from Y onto the range space of I. Since P is a linear operator and $A_{LS} = SI^{-1}P$, the algorithm A_{LS} is linear. Furthermore, it is not difficult to observe that A_{LS} is a correct algorithm.

The following very simply lemma will be used in the proofs of the main results of this section.

Lemma 1. *If Y is a Hilbert space then the element x_y defined in* (35) *is a symmetry center of $T(y)$ for any $y \in Y_0$, i.e.,*

$$\text{if } x \in T(y) \text{ then } x_1 = 2x_y - x \in T(y) \quad (37)$$

for any $x \in X$.

Proof. Let us observe that

$$\| Ix - y \|_H^2 = \| y \|_H^2 - \| Ix_y \|_H^2 + \| Ix - Ix_y \|_H^2. \quad (38)$$

Thus, $x \in T(y)$ iff

$$\| Ix - Ix_y \|_H^2 \leqslant \varepsilon^2 - \| y \|_H^2 + \| Ix_y \|_H^2. \quad (39)$$

Since

$$\| Ix - Ix_y \|_H = \| Ix_1 - Ix_y \|_H, \quad (40)$$

we have, due to our last observation, that

$$x \in T(y) \quad \text{iff} \quad x_1 \in T(y), \quad (41)$$

which means that x_y is a symmetry center of $T(y)$. □

In the following theorems we show optimality properties of least-squares algorithms assuming a Hilbert norm in Y. This assumption is discussed later in Remark 1.

Theorem 3. *If Y is a Hilbert space then A_{LS} is X-strongly optimal in the class $\mathscr{A}^c$ of correct algorithms.*

Proof. We have already noticed that $A_{LS} \in \mathscr{A}^c$. Let us now consider the error $E_1(A_{LS}, x)$. From the definition of A_{LS},

$$E_1(A_{LS}, x) = \sup_{y: \| y - Ix \|_H \leqslant \varepsilon} \| Sx - A_{LS}(y) \| = \sup_{y: \| y - Ix \|_H \leqslant \varepsilon} \| Sx - Sx_y \|. \quad (42)$$

We now show that if $\| y - Ix \|_H \leqslant \varepsilon$, i.e., $x \in T(y)$, then

$$\| Ix - Ix_y \|_H \leqslant \varepsilon.$$

Let $x_1 = 2x_y - x$. Since x_y is a symmetry center of $T(y)$ (see Lemma 1), we have that $x_1 \in T(y)$ and therefore

$$\| Ix - Ix_y \|_H = \tfrac{1}{2} \| Ix - Ix_1 \|_H \leqslant \tfrac{1}{2}(\| Ix - y \|_H + \| Ix_1 - y \|_H) \leqslant \varepsilon \tag{43}$$

which is the desired inequality. Using this fact in (42) we obtain that

$$E_1(A_{LS}, x) \leqslant \sup_{\hat{x}: \| I\hat{x} - Ix \|_H \leqslant \varepsilon} \| Sx - S\hat{x} \|. \tag{44}$$

From the other side, for any correct algorithm A we have

$$E_1(A, x) \geqslant \sup_{\hat{x}: \| I\hat{x} - Ix \|_H \leqslant \varepsilon} \| Sx - A(I\hat{x}) \| = \sup_{\hat{x}: \| I\hat{x} - Ix \|_H \leqslant \varepsilon} \| Sx - S\hat{x} \|. \tag{45}$$

Using now (44), we obtain that

$$E_1(A_{LS}, x) \leqslant E_1(A, x) \quad \forall A \in \mathscr{A}^c, \forall x \in X, \tag{46}$$

which proves the theorem. □

Theorem 4. *If Y is a Hilbert space, then A_{LS} is a central algorithm and therefore it is Y-strongly optimal in the class of all algorithms* (4).

Proof. We have to show that Sx_y is a center of $U(y)$ for $y \in Y_0$, i.e., that (see (19))

$$\sup_{x \in T(y)} \| Sx - Sx_y \| = \inf_{z \in Z} \sup_{x \in T(y)} \| Sx - z \|. \tag{47}$$

Let $z \in Z$ and $x \in T(y)$. Since x_y is a symmetry center of $T(y)$ (see Lemma 1), we have that $x_1 = 2x_y - x \in T(y)$. We now show that one of the following inequalities must hold:

$$\| Sx - Sx_y \| \leqslant \| Sx - z \| \tag{48}$$

or

$$\| Sx - Sx_y \| \leqslant \| Sx_1 - z \|. \tag{49}$$

Indeed, if both (48) and (49) are not satisfied for some $x \in T(y)$ and $z \in Z$ then

$$\| Sx - Sx_1 \| \leqslant \| Sx - z \| + \| Sx_1 - z \| < 2 \| Sx - Sx_y \|. \tag{50}$$

This is a contradiction, since $x - x_1 = 2(x - x_y)$ and therefore

$$\| Sx - Sx_1 \| = 2 \| Sx - Sx_y \|. \tag{51}$$

Since one of the inequalities (48) and (49) holds, we can write that

$$\| Sx - Sx_y \| \leqslant \sup_{x \in T(y)} \| Sx - z \| \quad \forall z \in Z, \forall x \in T(y) \tag{52}$$

This yields that

$$\sup_{x \in T(y)} \| Sx - Sx_y \| \leqslant \inf_{z \in Z} \sup_{x \in T(y)} \| Sx - z \|. \tag{53}$$

Since the opposite inequality is obvious, we obtain (47). This means that A_{LS} is a central algorithm. The Y-strong optimality of A_{LS} follows from Theorem 1. □

Remark 1. The assumption that Y is a Hilbert space is used in the proofs of Theorems 3 and 4 only to apply Lemma 1, i.e., to show that x_y given by (35) is a symmetry center of $T(y)$ for $y \in Y_0$. Suppose that Y is not a Hilbert space, but the set $T(y)$ has a symmetry center x_y for any $y \in Y_0$.

Then the conclusions of Theorems 3 and 4 remain valid for the algorithm A_{LS} defined by (36). Figure 2 shows the geometrical computation of $A_{LS}(y) \equiv A_c(y)$ for the same setting introduced in Example 1 with the only difference of an l_2 (instead of l_∞) norm in the space Y. It can

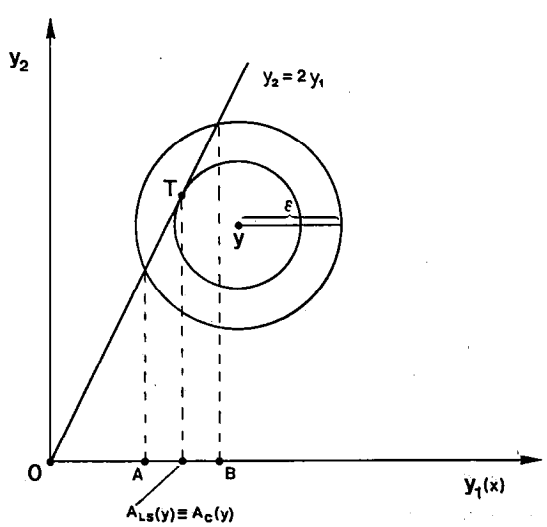

Fig. 2. The least-squares algorithm for l_2 norm in the space Y.

be observed that the least-squares solution is indeed a central algorithm, due to the fact that it takes the symmetry center of the set $T(y)$.

In the next section we address the question if the optimality properties states in Theorems 3 and 4 also hold for algorithms defined by (35) and (36) when the uncertainty of information is unknown but bounded with l_∞, l_1 or strictly convex norms in Y.

5. Are projection algorithms strongly optimal for other norms than Hilbert?

In this section we will examine cases in which the space Y is not necessarily equipped with a Hilbert norm.

As we have already mentioned at the end of the preceding section, we can generalize the least-squares algorithm by using definitions (35) and (36) with $\|\cdot\|_H$ replaced by some other norm, i.e., we can define an algorithm A_P by

$$\|Ix_y - y\| = \inf_{x \in X} \|Ix - y\| \tag{54}$$

and

$$A_P(y) = Sx_y \tag{55}$$

where the element x_y has been chosen according to some criterion among all elements satisfying (54). We shall call this algorithm a *projection algorithm*. An upper bound for the global error of such an algorithm can be easily computed as

$$E(A_P) \leqslant 2e(\varepsilon). \tag{56}$$

This result implies that if $\dim Z = 1$ or Z is equipped with an l_∞ norm, projection algorithms are 'nearly' optimal, in the sense that their global error is at most twice the intrinsic error of the problem.

It must be remarked that nonlinear estimation criteria such as for example least-absolute-values or minimax (see [1,2]), which are used in the literature as alternatives to least-squares criteria, actually correspond to the use of projection estimation algorithms in a space Y equipped with l_1 or l_∞ norms.

We ask if the X- and Y-strong optimality properties stated in Theorems 3 and 4 are preserved for projection algorithms in spaces equipped with non-Hilbert norms.

In the following example we consider an l_∞ norm in the space Y.

Example 3. We consider the same problem setting as in Examples 1 and 2. In particular, we assume that the space Y is equipped with l_∞ norm.

First we compute the projection algorithm $A_P(y)$. The determination of x_y in (54) is a minimization problem which can be solved geometrically by looking for the value of r for which the ball

$$\{\bar{y} \in \mathbb{R}^2 : \|\bar{y} - y\| \leqslant r\}$$

and the line $y_2 = 2y_1$, i.e., the range space of I, have a tangency point T (see Figure 1). Short computations lead to the following result:

$$Ix_y = \left[\tfrac{1}{3}(y_1 + y_2),\ \tfrac{2}{3}(y_1 + y_2)\right], \tag{57}$$

$$x_y = \tfrac{1}{3}(y_1 + y_2). \tag{58}$$

Taking into account that S is identity we obtain

$$A_P(y) = \tfrac{1}{3}(y_1 + y_2). \tag{59}$$

We now pass to the computation of a central algorithm. Three different situations may arise depending on the position of y with respect to the line $y_2 = 2y_1$:

$$\text{(A)}\quad 2y_1 - 3\varepsilon \leqslant y_2 \leqslant 2y_1 - \varepsilon, \tag{60}$$

$$\text{(B)}\quad 2y_1 - \varepsilon \leqslant y_2 \leqslant 2y_1 + \varepsilon, \tag{61}$$

$$\text{(C)}\quad 2y_1 + \varepsilon \leqslant y_2 \leqslant 2y_1 + 3\varepsilon. \tag{62}$$

Figure 1 shows a geometric interpretation of case (A). For this case we have already computed a central algorithm in Section 3 (see (21)). Similar computations for cases (B) and (C) lead to the following result:

$$\text{(A)}\quad A_c(y) = \tfrac{1}{4}(2y_1 + y_2 - \varepsilon), \tag{21}$$

$$\text{(B)}\quad A_c(y) = \tfrac{1}{2}y_2, \tag{63}$$

$$\text{(C)}\quad A_c(y) = \tfrac{1}{4}(2y_1 + y_2 + \varepsilon). \tag{64}$$

A comparison of (59) with (21), (63) and (64) shows that in l_∞ norm, opposite to what happens in Hilbert norm, a projection estimator is not necessarily central. Since the central algorithm A_c is a unique Y-strongly optimal algorithm (see Figure 1), we see that A_P is not Y-strongly optimal.

To investigate X-strong optimality we now compute the X-local and global (see (6) and (23))

errors of the projection and central estimators.

Since A_P is linear and correct we have [5]

$$E_1(A_P, x) = \varepsilon \| A_P \| \quad \forall x \in X. \tag{65}$$

Using (59) in (65) leads to

$$E_1(A_P, x) = \varepsilon \sup_{\|\rho\|_\infty \leq 1} \| A_P \rho \| = \tfrac{2}{3}\varepsilon. \tag{66}$$

As the local error of A_P does not depend on x, the global error is given by

$$E(A_P) = \sup_{x \in X} E_1(A_P, x) = \tfrac{2}{3}\varepsilon. \tag{67}$$

Let us now pass to the computation of the X-local error for A_c in the case (A). By using (21) in (6) we obtain

$$E_1(A_c, x) = \sup_{\substack{y: \|y - Ix\|_\infty \leq \varepsilon \\ y_2 \leq 2y_1 - \varepsilon}} |x - \tfrac{1}{4}(2y_1 + y_2 - \varepsilon)|. \tag{68}$$

It is not difficult to solve the above problem finding that the objective function attains its maximal value either for $y = (x + \varepsilon, 2x + \varepsilon)$ or for $y = (x, 2x - \varepsilon)$ (see Figure 3). Then it follows that

$$E_1(A_c, x) = \tfrac{1}{2}\varepsilon. \tag{69}$$

The computation of the X-local errors of A_c for the cases (B) and (C) can be performed in a similar way and it results that

$$E_1(A_c, x) = \tfrac{1}{2}\varepsilon \quad \forall x \in X. \tag{70}$$

As a consequence, we have for the global error

$$E(A_c) = \tfrac{1}{2}\varepsilon. \tag{71}$$

A comparison of (67) and (71) shows that A_P is not a globally optimal algorithm when an l_∞ norm is used in the space Y. Thus, A_P cannot be X-strongly optimal.

In the next example we shall consider the case in which the space Y is equipped with the l_1 norm.

Example 4. We now consider another example of a frequently used norm in Y, namely the l_1 norm. It is assumed that the general problem setting is still the same as for Example 1.

Let us compute the projection estimator (see Figure 4). As before we look for the tangency

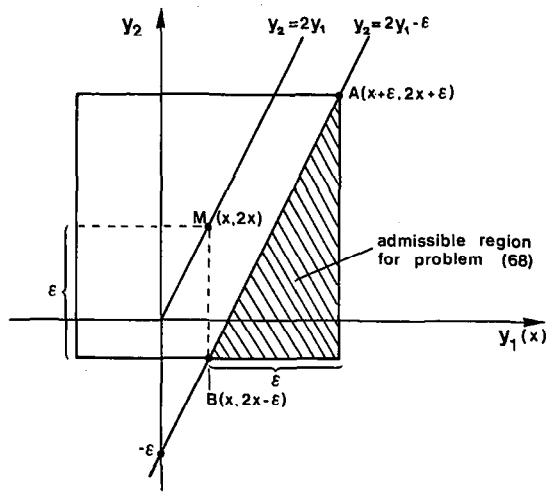

Fig. 3. A geometric sketch for the computation of the error of A_c.

point T. Straightforward computations lead to the following result:

$$A(y) = \tfrac{1}{2}y_2. \tag{72}$$

We observe now that for l_1 norm in Y, the set Y_0 (see (9)) is given by

$$Y_0 = \{ y \in Y : 2(y_1 - \varepsilon) \leq y_2 \leq 2(y_1 + \varepsilon) \}. \tag{73}$$

For what concerns the computation of central algorithms, also in this case we have three different possible cases:

$$\text{(A)} \quad 2(y_1 - \varepsilon) \leq y_2 \leq 2y_1 - \varepsilon, \tag{74}$$

$$\text{(B)} \quad 2y_1 - \varepsilon \leq y_2 \leq 2y_1 + \varepsilon, \tag{75}$$

$$\text{(C)} \quad 2y_1 + \varepsilon \leq y_2 \leq 2(y_1 + \varepsilon). \tag{76}$$

Figure 4 shows the situation in case (A). The segment EF represents both the sets $T(y)$ and $U(y)$. Thus, a central algorithm A_c takes the center of EF, i.e., in case (A),

$$A_c(y) = \tfrac{1}{3}(2y_2 - y_1 + \varepsilon). \tag{77}$$

Similar computations for the other cases lead to

$$\text{(B)} \quad A_c(y) = \tfrac{1}{3}(y_1 + y_2), \tag{78}$$

$$\text{(C)} \quad A_c(y) = \tfrac{1}{3}(2y_2 - y_1 - \varepsilon). \tag{79}$$

By comparing (77), (78) and (79) with (72), it easily follows that also for l_1 norm the projection estimator A_P is not necessarily central, i.e., it is not Y-strongly optimal. To study the optimality and X-strong optimality of A_P we compute the

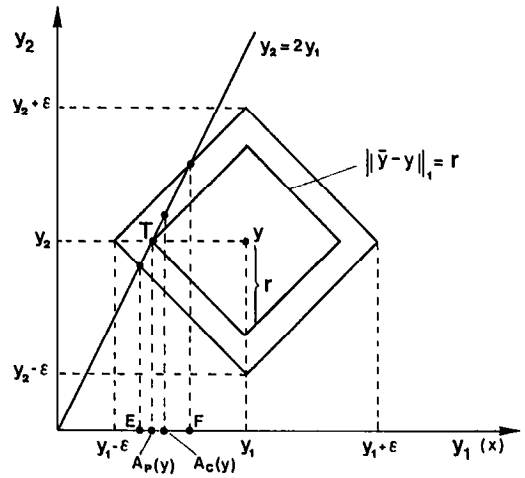

Fig. 4. A geometric sketch for the computation of the central algorithm A_c and the projection algorithm A_P for l_1 norm in the space Y.

X-local error of A_P and the intrinsic error of the problem. The X-local error is given by

$$E_1(A_P, x) = \varepsilon \| A_P \| = \varepsilon \sup_{\|\rho\| \leq 1} |A_P \rho| = \tfrac{1}{2}\varepsilon. \quad (80)$$

Hence we have that the global error of A_P is equal to

$$E(A_P) = \sup_{x \in X} E_1(A_P, x) = \tfrac{1}{2}\varepsilon. \quad (81)$$

Let us now compute the intrinsic error of the problem which is given by (see (27) and (28))

$$E_e = e(\varepsilon) = \sup_{\|Ix\|_1 \leq \varepsilon} \|Sx\|. \quad (82)$$

Since $E(A_c) = E_e$ (see (25)) we have:

$$E(A_c) = \sup_{\substack{|y_1| + |y_2| \leq \varepsilon \\ y_2 = 2y_1}} |y_1| = \tfrac{1}{3}\varepsilon. \quad (83)$$

A comparison of (81) and (83) shows that for the norm l_1, A_P is not optimal, i.e., it cannot be X-strongly optimal.

The next example deals with a parameter estimation problem when a strictly convex norm is assumed in the space Y.

Example 5. In this example we refer to the same general setting as for Example 1, but we consider a particular strictly convex norm in Y.

We recall that the norm in Y is called strictly convex if

$$\text{if } y_1 \neq y_2,\ \|y_1\| = \|y_2\| = 1 \quad \text{then } \|y_1 + y_2\| < 2 \quad \forall y_1, y_2 \in Y. \quad (84)$$

It is to observe that the class of strictly convex norms includes Hilbert norms and for this reason one might wonder if the X-strong optimality and centrality properties of projection algorithms for Hilbert norms (see Theorems 3 and 4) are preserved for the more general class of strictly convex norms. We remark that the norms l_∞ and l_1 are not strictly convex.

We now introduce a strictly convex norm $\|\cdot\|_s$. Let

$$C = \{(y_1, y_2) : y_1^2 - 1 \leq y_2 \leq 1 - y_1^2\}. \quad (85)$$

The convex bounded set C has a center of symmetry at 0 and it is absorbing, i.e., for any $y \in Y$, $\alpha y \in C$ for some $\alpha > 0$. We define $\|\cdot\|_s$ using the Minkowski functional [11]

$$\|y\|_s = \inf\{\alpha^{-1} : \alpha y \in C,\ \alpha > 0\}. \quad (86)$$

It is not difficult to verify that for $r > 0$,

$$\|y\|_s \leq r \quad \text{iff} \quad y \in K(0, r), \quad (87)$$

where

$$K(0, r) = \left\{(y_1, y_2) : \frac{1}{r}(y_1^2 - r^2) \leq y_2 \leq -\frac{1}{r}(y_1^2 - r^2)\right\}. \quad (88)$$

We note that the norm $\|\cdot\|_s$ defined above is strictly convex (see Figure 5).

Let us now compute the projection estimator A_P. Given $y \in Y$, we look for the radius r such that $K(y, r)$ is tangent to the range space of I (see Figure 5). It is easy to obtain that

$$A_P(y) = \tfrac{1}{2} y_2. \quad (89)$$

We shall now show that A_P is neither Y- nor X-strongly optimal. Let us compute the X-local error of A_P. We again use (65) obtaining

$$E_1(A_P, x) = \varepsilon \| A_P \| = \varepsilon \sup_{\|\rho\|_s \leq 1} |A_P \rho| = \tfrac{1}{2}\varepsilon. \quad (90)$$

As the X-local error is constant, we obtain that the global error is

$$E(A_P) = \tfrac{1}{2}\varepsilon. \quad (91)$$

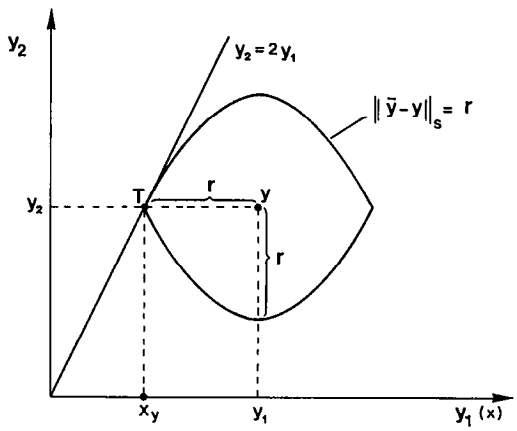

Fig. 5. The projection algorithm in the space Y with a strictly convex norm.

The intrinsic error of the problem is given by (see (27) and Figure 6)

$$E_\varepsilon = e(\varepsilon) = \sup_{\substack{\|y\|_s \leq \varepsilon \\ y_2 = 2y_1}} |y_1| = (\sqrt{2} - 1)\varepsilon. \tag{92}$$

A comparison of (91) and (92) shows that A_P is not (globally) optimal and hence it is neither X- nor Y-strongly optimal. As a consequence, Theorems 3 and 4 of the preceding section cannot be extended to be whole class of strictly convex norms.

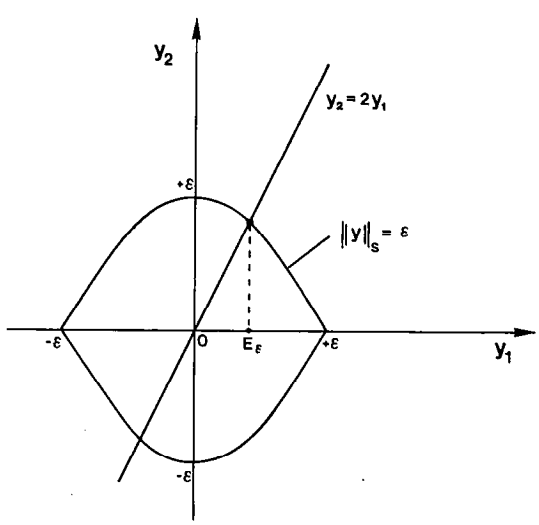

Fig. 6. A geometric sketch for the computation of the intrinsic error for a strictly convex norm in the space Y.

6. Conclusions

Optimality properties of projection algorithms frequently used in problems of system identification have been studied. We have assumed that the measurement noise is deterministic, i.e., unknown but bounded in a suitable norm. The main results obtained refer to least-squares algorithms corresponding to the use of a Hilbert norm. Optimality properties shown in Section 4 explain some of the reasons why least-squares algorithms because satisfactorily also in practical situations where standard statistical assumptions are not satisfied. As pointed out in Section 5 these robust optimality properties do not hold for least-absolute-values or minimax algorithms.

It could be of interest in future work to investigate relations between the unknown but bounded and the statistical approaches with the aim of better understanding the optimality properties of least-squares algorithms in both contexts.

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